RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001: 2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Mechanical Engineering
Academic Year 2020 – 2021

Minutes of Meeting – 3rd Program Assessment & Quality Improvement Committee (PA&QIC)

Date of Meeting : 19.07.2021 (Monday)
Time of Meeting : 3:00 PM to 4:30 PM
Reference : RIT/Mech./PA&QIC/03
Google Meet Link : https://meet.google.com/xtn-qshk-oeb

Members Present:

Sl. No.	Name of the Faculty Member	Designation	PAQIC
1.	Dr.S.Rajakarunakaran	Professor and Head / Mechanical	Convener
2.	Mr.M.Ashok Kumar	AP(SG) / Mechanical	Member
3.	Mr.S.Maharajan	Assistant Professor / Mechanical	Member
4.	Mr.R.Prabhakaran	Assistant Professor / Mechanical	Member
5.	Mrs.R.Kalaimani	Assistant Professor / Civil	Member
6.	Mr.C.A.Yogaraja	Assistant Professor / CSE	Member
7.	Mr.A.Arun Kumar	Assistant Professor (SG) / EEE	Member
8.	Mrs.R.Ramalakshmi	Assistant Professor (SG) / ECE	Member
9.	Dr.G.Selvaraj	Assistant Professor / Maths	Member
10.	Dr.P.Suresh Kumar	Associate Professor / Mechanical	Representative
11.	Dr.V.Sivakumar	Associate Professor / Mechanical	Representative
12.	Dr.S.Godwin Barnabas	Associate Professor / Mechanical	Representative
13.	Mr.J.Jerold John Britto	AP(SG) / Mechanical	Representative
14.	Mr.M.Sivagaminathan alias Balaji	Assistant Professor (SG)	Representative
15.	Mr.S.Valai Ganesh	AP / Mechanical	Representative
16.	Mr.C.Gururaj	AP(SG) / Mechanical	Representative
17.	Mr.J.Jabinth	AP / Mechanical	Representative
18.	Mr.L.Karthikeyan	AP / Mechanical	Representative

27.8.21

Sl. No.	Name of the Faculty Member	Designation	PAQIC
19.	Mr.G.Prabu ram	AP(SG) / Mechanical	Representative
20.	Mr.M.Santhana Maruthu Pandian	AP / Mechanical	Representative
21.	Mr.R.Arun Kumar	AP / Mechanical	Representative
22.	Mr.R.Venkatesh	AP / Mechanical	Representative
23.	Mr.N.L.Sujin	AP / Mechanical	Representative
24.	Mr.T.Selva Sundar	AP / Mechanical	Representative
25.	Mr.M.Ramar	AP / Mechanical	Representative
26.	Mr.S. Kathiravan	AP / Mechanical	Representative
27.	Mr.P.Pavithran	AP / Mechanical	Representative

The agenda of the meeting

1. Review of Previous meeting – 2nd PA&QIC
2. Result analysis of 2019-2020 Even Semester & 2020-2021 Odd Semester
3. Program Curriculum and Teaching and Learning Process
4. Attainment of COs, POs, PSOs with program effectiveness (2019-2020)
5. Institute – Industry Interactions and its impact
6. Training and Placement Progress & Career Guidance Cell activities
7. Professional Societies and Technical Association: Participants and its inference
8. Internship & In-Plant Training
9. EDC, IIC and NISP Progress
10. Faculty participations / performance
11. Laboratory utilization with respect to OBE
12. Alumni Association Interactions
13. Continuous Improvements
14. Stakeholders Feedback and analysis: Students/Parents/Alumni/Recruiters
15. Any other matters

1. REVIEW OF PREVIOUS MEETING (2ND PA&QIC)

Dr.P.Sureshkumar, ASCP/Mech. presented the progress and action taken for the suggestions given in 2nd PA&QIC meeting held on 21.11.2020.

Item	Suggestions	Action taken
Teaching & Learning Practiced	Ensure the effective use of LMS	Faculty submitted the course Analytics during DRM
	Update the innovative practices for the year 2020-2021	Updated in the file as well as website
	Maintain the transparent evaluation and Rubrics	Rubrics has been developed and circulated to student before the evaluation
	Different LMS Platform	Different LMS platform used by the faculty members
Assessment Methods, Attainment of CO, PO, PSO	Set the higher target level	Implemented
	Reason for decreasing trend in CO-Attainment	Analyze done and the results present in this meeting
Virtual Lab Utilization	To collect the Feedback and outcome from student	Collected the feedback and analyzed the outcome
T&P	Motivate the virtual internship	25 student have completed
	Software training for RIT-Harita	Training conducted and placed the students
	GD conducted along with English faculty	English faculty included in GD
Internship and IPT	Improve the participation	IPT: 09 (19-20), 20 (20-21) Intern: 49 (19-20), 35 (20-21)
	Periodically collect the feedback from organization	Collected the feedback
Professional Societies and Association	Summary table for benefitted students	Discussed in this meeting
Faculty Participation	Comparative Analysis has to take for two years	Comparative analysis made by the respective criterion coordinator
Project Proposal/ Research Publication	Specify the classification i.e SCI and Scopus Journal	Separately mentioned SCI and Scopus journal
Stakeholder feedback	Training program need for usage of LMS	Special program were conducted for canvas

2. RESULT ANALYSIS OF 2019-2020 EVEN SEMESTER AND 2020-2021 ODD SEMESTER

Mr.T.Selva Sundar, AP/Mech. presented the result analysis for April /May 2020 and Nov /Dec 2020.

- Micro level analysis were made for failure students and reasons are listed out
- The subject wise grade analysis was done to find the impact of GPA for all subjects

- As per the direction of Govt. of Tamil Nadu, Anna University conducted the re exam for the interested students those who want to get pass and improve their grade. Many students attended re exam in order to improve their CGPA and all the arrear students also attended Re exam to get pass mark.

Reason for Failure:

- Few students faced login issues due to their poor network connectivity.
- Since it is MCQ type questions, slow learners found difficulties in solving problems which were asked in MCQ.
- Students spent more time to solve analytical questions that were asked in MCQ format, Due to lack of time management, they didn't answer all the questions in time.
- In ME8072 Renewable Sources of Energy & ME8073 Unconventional Machining Processes (First exam for VII semester), 90% of the failed students reported that they faced login issues during the exam. Some students complained that their test got submitted due network problem and hence they did not attend remaining questions.

Suggestion Given by members:

- In some subjects average grade point value is below 7 GPA. PAQIC Members suggested collecting and analyzing the reasons given by the subject handling faculty and take necessary actions for the betterment.
- Students were advised to attend their exam from the place where the network strength is good.

3. PROGRAM CURICULUM, TEACHING AND LEARNING PROCESS

Presenter: Mr.G.Prabu Ram, AP (Sr.Gr)/Mechanical and Mr.R.Arun Kumar, AP/Mechanical

Mr.G.Prabu Ram, AP (Sr.Gr)/Mechanical briefed the Teaching learning followed in the department as follows:

A. PROGRAM CURICULUM:

Curriculum Gap Identification:

The curriculum gap identification process was explained. It was explained that the curriculum gap is identified based on feedback from Industry expert, Academic expert, Parents, Students and Alumni. For the same the department courses were divided in to three broad domains and domain coordinators were nominated.

The detail of the same is listed below.

Sl. No.	Domain	Coordinator	Members
1.	Thermal	Mr.M.Ashok Kumar, AP(SG)	Dr.V.Sivakumar, ASCP Mr.M.Sivagaminathan alias Balaji, AP(SG) Mr.R.Arun Kumar, AP Mr.N.L.Sujin, AP
2.	Design	Mr.J.Jerold John Britto, AP(SG)	Mr.C.Gururaj, AP(SG) Mr.G.Prabu ram, AP(SG) Mr.S.Valai Ganesh, AP Mr.J.Jabinth, AP Mr.L.Karthikeyan, AP Mr.R.Venkatesh, AP Mr.R.Prabhakaran, AP Mr.T.Selva Sundar, AP
3.	Manufacturing	Mr.S.Maharajan, AP	Dr.S.Rajakarunakaran, Professor and Head Dr.P.Suresh Kumar, ASCP Dr.S.Godwin Barnabas, ASCP

Sl. No.	Domain	Coordinator	Members
			Mr.M.Lakshmanan, AP(SG) Mr.M.Santhana Maruthu Pandian, AP Mr.M.Ramar, AP Mr.S.Kathiravan, AP Mr.P.Pavithran, AP

The domain wise meeting was conducted by the domain in charges and the faculty members were allotted with courses and to collect the curriculum gap suggestions from different stake holders such as parent, students, industrial experts, professional society representatives, peer groups. The curriculum was circulated among stake holders in third week of July, 2021. The gaps were suggested by the experts and consolidated during last week of July. The identified gaps were submitted to Anna University on 27.07.2021. The gap identification was course specific as well as curriculum specific.

The domain committee reviewed the suggestions and the curriculum gaps were sent to University Centre for Academic Courses in following aspects:

- Addition of curricular content.
- References to be included in the existing curriculum as well as for the proposed curriculum.
- Rearrangement of courses in the curriculum to meet pre-requisites.
- Suggestion of new courses to cater the needs of industrial demands.

Based on the suggestions from experts, the following courses are planned to be offered as Value Added Course in the AY 2021 – 22 and the approval request for the same was sent to Anna University Centre for Academic Courses on 27.07.2021. The lists of courses are:

1. IoT and AR Applications in Mechanical Engineering
2. Modern Fabrication Techniques of Plastics
3. Chatbot Design for Industrial Applications

The curriculum content for the above said courses includes references, pre-requisites, objectives, outcomes, assessment methods along with detailed delivery plan/schedule.

Based on the input from criteria 7 (Continuous Improvement), it is decided to continue the following two Value Added Courses.

MVA005 – Smart Materials and Structures

MVA006 – Green Energy Technologies and Management

Note: It was presented that two new forms to be introduced in AY 2021 – 22 for strengthening the curriculum gap identification process.

- Student's feedback on curriculum
- Faculty's feedback on curriculum

Content Beyond Syllabus Plan:

It was decided in all the domain meetings to address the curriculum gap identified by planning content beyond syllabus in the respective courses and the same has to be mapped with corresponding POs, PSOs. Faculty members will be following an assessment tool for the topics covered through content beyond syllabus and the same will be considered for attainment calculation.

B. TEACHING LEARNING PROCESS:

Innovative Practices:

The innovative practices followed in different courses were presented by Mr.G.Prabu ram. Few sample practices are listed below:

Sl. No.	Course Code & Title	Practice followed	Topic
1.	GE8075 Intellectual Property Rights	Information sharing platform	IPR Infringement Case Studies
2.		Quizziz online poll game	End of Unit IV
3.		Online course assignment	Patent informatics

Sl. No.	Course Code & Title	Practice followed	Topic
4.		Guest Lecture	Patent Search and applying digital signature for Patent filing
5		Feedback based learning	Patent Drafting
6.	CE8395 Strength of Materials for Mechanical Engineers	Use of Simulation Software (Beamguru)	Shear Force and Bending Moment Diagram
7.	ME8692 – Finite Element Analysis	T2P Description using software	Beam FEA using CREO
8.	PR8592 – Welding Technology	Cross Word Puzzle	Unit I overview
9.		Virtual welding using mobile app (Welducation)	Types of welding
10.		Connection with welding	Welding basics
11.	ME8693 Heat and Mass Transfer	Problem solving using MAT Lab simulation	Forced convection problem

Based on the input from criteria 7 (Continuous Improvement) and feedback through course exit survey it is proposed to follow online simulation tools like MAT Lab, Software, etc. for making the students to understand the concepts.

Methods to support slow and bright learners:

It was presented that the following steps were taken for the slow learners and bright learners in the department.

Slow learners: Special classes to face the end semester examination conducted by Anna University

Bright Students:

- GATE coaching is given for the students to face the technical competitive exams.
- Motivating students to pursue online courses to strengthen their area of interest.

It was suggested that the effectiveness of special classes can be presented. The outcome of special class can be measured and used for Course outcome attainment calculation.

Quality of internal assessment question papers:

It was presented that the quality of question papers for internal assessment tests are assessed in two levels:

- Internal review
- Centralized review

The respective domain coordinators will be verifying the quality of question papers by checking the adherence to standards including question quality, ISO standards of institution, course outcomes, Blooms taxonomy level and Anna University norms in internal review mechanism.

In external review mechanism, the random samples of question papers will be verified by the domain experts by checking the above said adherence and questions quality.

Quality of Project:

It was presented as follow for the project quality and review mechanism.

- Students are asked to fill the project form along with tentative title of the project and willing supervisor.
- Based on the chosen area of project, the students are allotted with supervisors with expertise in the proposed area.
- Students will undergo three internal review processes for which the Review Committee is formulated in respective departments.
- The second review for the students project will involve an invited industrial expert to suggest the students for further improvisation in the project

- After completion of the project the outcome of student's projects are categorized as follows:
  - Patents
  - Publications
  - Conference presentations
  - Project competitions

Internship and its impact on Program Outcomes:

It was presented as follow for the internship initiatives:

- Due to pandemic crisis, students were encouraged to undergo virtual internship. The details of student internship were presented and the same is listed below in table.

Sl. No.	Name of the Company	No. of students
1.	Intech olympiad online summer internship (A Four Technologies)	13
2.	Ezenith Education	01
3.	Uniq Technologies	01
4.	Internship studio	01
5.	Cognitech	02
6.	Xyma Analytics, IIT-M research Park, Chennai	01
7.	Ramco Cements, RR Nagar, Virudhunagar	03
8.	Rajapalayam Mills Limited, Rajapalayam	01
9.	Summits Hygronics Pvt Limited	02
10.	SB Engineering, Coimbatore	01
11.	Rane Engine Valve Limited, Pudukottai	02
12.	National Instruments	04

- The department consultative committee was constituted and the committee will assess the students' outcome through internship
- Students submit their internship report and they were asked present the same to the consultative committee.
- An exclusive rubrics is deigned to evaluate the performance of the students.
- It was suggested to correlate the internship activities with suitable POs and PSOs so that the same can be used for attainment calculation

Suggestions given by the members

1. Separate discussion may be given on Outcome of Value added courses and Innovative practices.
2. Try to measure the outcome of slow learners after conducting special classes
3. Modern tool usage can be implemented for all subjects.
4. Few implementation proofs after getting feedback from students can be presented
5. Internship assessment can be considered for attainment calculation.

4. ATTAINMENT OF COs, POs, PSOs WITH PROGRAM EFFECTIVENESS (2019-20)

Presenter: Mr.J.Jabinth, AP/Mech.

The points presented in the 3rd PAQIC meeting are as follows,

a. Assessment methods practiced in the last academic year.

Direct	Indirect
Internal Assessment Test 1,2,3, Assignment, Quiz, Course Exit Survey	Alumni Exit survey, Program Exit survey

b. The details of courses considered for attainment calculations are,

No. of Course Outcomes	ODD:	EVEN:
Project	1	1
Elective Course	4	6
Practical Course	10	11
Theory Course	26	24

c. The details of course outcomes considered for attainment calculations,

No. of Courses and their levels	L1	L2	L3	L4
ODD	19	79	107	13
EVEN	21	68	78	09

d. Details of co-correlations with POs & PSOs

ODD

PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
36	34	35	37	22	18	14	30	17	36	1	34

ODD

PSO1	PSO2	PSO3	PSO4
18	26	19	16

EVEN

PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
29	28	28	26	20	14	11	29	15	32	6	30

EVEN

PSO1	PSO2	PSO3	PSO4
16	18	16	9

e. Details of Target Comparison

Batch	Target Attainment
2016-2021 ($AGPA_1$) – R13	$\left(\frac{(AGPA_1 + AGPA_2 + AGPA_3)}{2}\right) \times 0.3$
2017-2022 ($AGPA_1$) – R17	1.5 for Theory Courses & 2 for Laboratory Courses/Projects
2018-2023 ($AGPA_2$) – R17	($AGPA_1 \times 0.3$)

Batch	Target Attainment
2019-2024 (AGPA₃) – R17	Maximum of $\left(\left(\frac{(AGPA_1 + AGPA_2)}{2}\right) \times 0.3, (AGPA_1 \times 0.3)\right)$

Course Code	Course Name	Target			
I Semester		2013-17	2014-18	2015-19	2016-19
C101	Technical English – I	1.5	1.95	1.98	1.96
C102	Mathematics – I	1.5	2.51	2.2	2.09
C103	Engineering Physics – I	1.5	1.99	2.09	2.08
C104	Engineering Chemistry – I	1.5	1.97	2.12	2.14
C105	Computer Programming	1.5	2.05	2.06	2.04
C106	Engineering Graphics	1.5	2.59	2.39	2.22
C107(L)	Computer Practices Laboratory	2	2.77	2.79	2.75
C108(L)	Engineering Practices Laboratory	2	2.67	2.69	2.7
C109(L)	Physics and Chemistry Laboratory - I	2	2.66	2.64	2.67
II Semester		2013-17	2014-18	2015-19	2016-120
C110	Technical English – II	1.5	1.86	2	2.09
C111	Mathematics – II	1.5	2.07	2.02	1.96
C112	Engineering Physics – II	1.5	2.22	2.08	1.98
C113	Engineering Chemistry – II	1.5	2.02	1.99	2.02
C114	Basic Electrical and Electronics Engineering	1.5	1.76	1.86	1.91
C115	Engineering Mechanics	1.5	2	1.96	1.95
C116(L)	Computer Aided Drafting and Modeling Laboratory	2	2.87	2.83	2.83
C117(L)	Physics and Chemistry Laboratory - I	2	2.69	2.67	2.72
III Semester		2013-17	2014-18	2015-19	2016-120
C201	Transforms and Partial Differential Equations	1.5	2.15	1.91	1.86
C202	Strength of Materials	1.5	1.95	2.03	2
C203	Engineering Thermodynamics	1.5	1.77	1.5	1.7
C204	Fluid Mechanics and Machinery	1.5	1.74	1.89	1.96
C205	Manufacturing Technology - I	1.5	2.07	2.1	2.08
C206	Electrical Drives and Controls	1.5	2.06	2.8	1.98
C207(L)	Manufacturing Technology Laboratory - I	2	2.78	2.78	2.77
C208(L)	Fluid Mechanics and Machinery Laboratory	2	2.66	2.71	2.36
C209(L)	Electrical Engineering Laboratory	2	2.82	2.8	2.76
IV Semester		2013-17	2014-18	2015-19	2016-120
C210	Statistics and Numerical Methods	1.5	2.83	2.12	2.07
C211	Kinematics of Machinery	1.5	2.03	1.97	1.93
C212	Manufacturing Technology– II	1.5	2.13	2.08	2.06
C213	Engineering Materials and Metallurgy	1.5	2.09	2.11	2.07
C214	Environmental Science and Engineering	1.5	2.02	2.02	2
C215	Thermal Engineering	1.5	1.94	1.93	1.87

C216(L)	Manufacturing Technology Laboratory–II	2	2.84	2.83	2.79
C217(L)	Thermal Engineering Laboratory - I	2	2.79	2.75	2.72
C218(L)	Strength of Materials Laboratory	2	2.83	2.84	2.84
	V Semester	**2013-17**	**2014-18**	**2015-19**	**2016-120**
C301	Computer Aided Design	1.5	2.14	2.15	2.11
C302	Heat and Mass Transfer	1.5	2.05	2.06	2.02
C303	Design of Machine Elements	1.5	2.03	1.98	1.95
C304	Metrology and Measurements	1.5	2.07	2.16	2.13
C305	Dynamics of Machines	1.5	2.18	2.2	2.06
C306	Professional Ethics in Engineering	1.5	2.08	2.11	2.07
C307(L)	Dynamics Laboratory	2	2.79	2.79	2.4
C308(L)	Thermal Engineering Laboratory-II	2	2.75	2.76	2.36
C309(L)	Metrology and Measurements Laboratory	2	2.83	2.8	2.39
	VI Semester	**2013-17**	**2014-18**	**2015-19**	**2016-120**
C310	Design of Transmission Systems	1.5	2.17	2.1	2.08
C311	Principles of Management	1.5	2.2	2.17	2.04
C312	Automobile Engineering	1.5	2.19	2.21	2.15
C313	Finite Element Analysis	1.5	2.24	2.13	1.89
C314	Gas Dynamics and Jet Propulsion	1.5	2.23	2.06	1.94
C315	Renewable Source of Energy	1.5	2.29	2.18	2.22
C316	Unconventional Machining Process	1.5	2.29	2.18	2.22
C317(L)	C.A.D. / C.A.M. Laboratory	2	2.97	2.9	2.78
C318(L)	Design and Fabrication Project	2	2.9	2.92	2.86
C319(L)	Communication and Soft Skills-	2	2.42	2.49	2.49
	VII Semester	**2013-17**	**2014-18**	**2015-19**	**2016-120**
C401	Power Plant Engineering	1.5	2.34	2.18	2.12
C402	Mechatronics	1.5	2.37	2.07	2.02
C403	Computer Integrated Manufacturing Systems	1.5	2.23	2.12	2.11
C404	Total Quality Management	1.5	2.24	2.08	2.03
C405	Process Planning and Cost Estimation	1.5	2.41	2.21	2.12
C406	Welding Technology	_	_	1.5	2.08
C407	Energy Conservation and Management	1.5	2.41	2.21	2.08
C408	Maintenance Engineering	1.5	2.25	2.06	2.06
C409(L)	Simulation and Analysis Laboratory	2	2.99	2.91	2.84
C410(L)	Mechatronics Laboratory	2	2.95	2.81	2.77
C411(L)	Comprehension	2	2.48	2.36	2.57
	VIII Semester	**2013-17**	**2014-18**	**2015-19**	**2016-120**
C412	Engineering Economics	1.5	2.16	2.05	2.02
C413	Production Planning and Control	1.5	2.18	2.24	2.13
C414	Entrepreneurship Development	_	1.5	2.24	2.13
C415	Non Destructive Testing and Materials	1.5	2.21	2.06	1.99
C416	Advanced IC Engines	1.5	_	2.21	1.99
C417(L)	Project Work	2	2.87	2.92	2.91

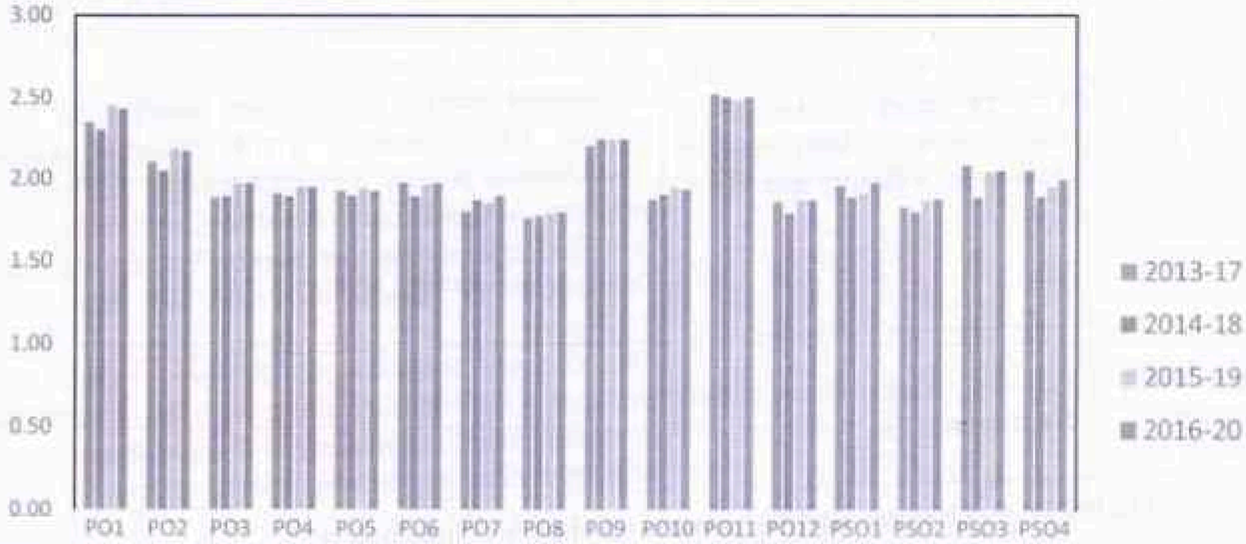
Details of CO Not Attainment subjects

Sl.No.	Code	Name	Target	CO	Status
1	C102	Engineering Mathematics I	1.8	1.01	Not Attained
2	C103	Engineering Physics	1.8	1.45	Not Attained
3	C104	Engineering Chemistry	2.11	0.27	Not Attained
4	C105	Problem Solving and Python Programming	2	0.46	Not Attained
5	C109	Technical English	2.01	1.89	Not Attained
6	C201	Transforms and Partial Differential Equations	1.8	0.52	Not Attained
7	C202	Engineering Thermodynamics	1.8	1.16	Not Attained
8	C203	Fluid Mechanics and Machinery	1.8	1.05	Not Attained

Highly Attained subjects (>40% improvement compared to target):

Code	Subject Name	%
C419	Additive Manufacturing	48
C415	Non Destructive Testing and Materials	48
C303	Design of Machine Elements	45
C408	Maintenance Engineering	43
C406	Welding Technology	43
C202	Strength of Materials	42
C404	Total Quality Management	42

f. Details of PO Attainment



	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
2013-17	2.35	2.10	1.88	1.91	1.93	1.97	1.81	1.76	2.21	1.87	2.52	1.86	1.96	1.83	2.08	2.05
2014-18	2.31	2.05	1.89	1.89	1.90	1.90	1.87	1.77	2.24	1.90	2.50	1.79	1.89	1.80	1.89	1.89
2015-19	2.46	2.18	1.97	1.95	1.94	1.96	1.86	1.79	2.24	1.95	2.48	1.87	1.92	1.87	2.04	1.95
2016-20	2.44	2.17	1.97	1.95	1.93	1.97	1.90	1.79	2.24	1.94	2.50	1.87	1.98	1.88	2.05	1.99

<u>Highly attained POs</u>: PO7 & PSO1

5. INDUSTRY INSTITUTE INTERACTIONS AND ITS IMPACT

Presenter: Mr.R.Venkatesh, AP/Mechanical. and Dr.S.Godwin Barnabas, ASCP/Mechanical

Dr.S.Godwin Barnabas, ASCP/Mechanical discussed the following activities:

1.Harita Techserv Ltd, Chennai.

S.No.	Software Used	No.of Students Trained	No .of Students Placed
1.	CATIA Software	17	3

2. Ramco Group of Textile Division, Rajapalayam.

S.No.	Outcomes	Details
1.	Final year project completion	S.V.Vignesh final year Mechanical Engineering student completed Industry project under the guidance of Mr.Abhinav Kumar,Trainee, Ramco Group of Textile Division, Rajapalayam
2.	Paper presentation in International conference	S.V.Vignesh final year mechanical student presented paper titled "Implementation of KAIZEN concept in Textile Industry in international conference conducted by Francis Xavier college of engineering, ICRAME March 30,2021.

3. Gowri House Metal Works, Rajapalayam.

S.No	Outcomes	Details
1.	Two Industry Institute Lecture series conducted by the Ramco Institute of Technology faculty members.	1. First lecture series were handled by Dr.P.Suresh Kumar ASCP/Mech and Dr.O.Senthil Kumar ASCP/Chemistry on a title "Heat treatment process" on 24.01.2021 from 09.30a.m to 3.30pm. 2. Second lecture series were handled by Dr.P.Suresh Kumar ASCP/Mech on a title Material properties and Dr.O.Senthil Kumar ASCP/Chemistry & Dr.M.Venkatesh perumal AP/Chemistry on a title "Lubricants and coolants" and final session was handled by Mr.M.Santhana Maruthu Pandian on AP/Mech on 28.03.2021 from 09.00a.m to 12.30pm.
2.	Industry visit cum proble discussion	Dr.S.Godwin Barnabas,ASCP/Mech , Dr.M. Venkatesh perumal AP/Chemistry along with the 3 year mechanical student visited the company to discuss about the possibility of oil recycling in manufacturing process.

4. MEDSBY Healthcare and Engineering Solutions Coimbatore.

S.No	Outcomes obtained	Outcomes Details
1.	Award achievement in national level	Received Best Collaborator award in National Level, Faculty and student training, Product developed using additive manufacturing
2.	Prizes secured by the students	With the support of Medsby our students secured 1st,2nd and 3rd prizes in 3D printing competition conducted by IITM PALS.

S.No	Outcomes obtained	Outcomes Details
3.	Consultant work to other organizations.	3D printing consultant activity carried out for the SACS MAVMM Engineering college for the amount worth of Rs 8,500/-

5. Ramco Cements Ltd.,Chennai.

S.No	Outcomes obtained	Outcomes Details
1.	Students Internship	Final Year and 3rd year Mechanical Engineering 08 students undergone internship in Jan-March 2021.
2.	Students Project Work	Two final year students completed project work in the title of 'Automation in Cement Bag Feeding in Electronic Rolling Packer'
3.	Paper Presentation	One student presented a paper titled 'Automation in Cement Bag Feeding in Electronic Rolling Packer', in the National Level Symposium on 'Recent Trends in VLSI Signal Processing, IoT, Communication & Embedded Systems-VSPICE 2021 conducted by Amirta College of Engineering and Technology on 3rd May 2021.

6. NI Systems Private Limited, Bangalore.

S.No	Outcomes obtained	Outcomes Details
1.	Students Training Programs	05 Students are get a training from NI Systems
2.	Students obtained certificates	Three final year students obtained CLAD certificate from NI Systems

7. Aravind Herbal, Rajapalayam.

S.No	Outcomes obtained	Outcomes Details
1.	Proposal Submission	1.Project proposal submitted by the mechanical department faculty members in DST under WMT theme titled "Dual benefits in using the waste tyre for waste water treatment and production of automotive seals and gaskets",on 30.11.2020. 2.Submitted proposal got screened for final presentation by DST expert members on 14.07.2021.
2.	Providing solutions to the Industry problems	Problems faced by the industry for capsule filling in automated manner is given as fabrication work to the students guided by the internal faculty members.

8. V-Invent Chemilab Private Limited, Rajapalayam.

- Working model has been developed in front of the V-Invent Proprietor. Three faculties and four students are involved in this activity.

S.No	Outcomes obtained	Outcomes Details
1.	Design of Plaster Panel	2D schematic sketch developed using AutoCAD software.
2.	Developed Plaster Panel	Model developed using Bullet screw and 3D Printed Plate head Techniques.

S.No	Outcomes obtained	Outcomes Details
3.	Automation of Painting Systems	Automation Concept planned and it has been proposed with dual nozzle systems.

9. Raja Nursery Garden, Rajapalayam

S.No	Outcomes obtained	Outcomes Details
1.	Forest App Development	1.Applied REAL FOREST mobile app to start-up to be a part of NIAM-Startup Agri-Business Incubation Programme with Registration Number: Reg//NIIAM// AGRIIBUSIINESS//2020//40. 2.Started Phase-I Work of REAL FOREST applications Module Zero portion.
2.	Proposal Submission	RIT and Raja Nursery Garden submitted proposal titled "Agro Real Forest Digitalization"under "Start-up Agri business incubation programme" worth of Rs.25,00,000/- to Centre for Innovation, Entrepreneurship and Skill Development, CCS National Institute of Agricultural Marketing

- **Suggestions given by the member and future plans to improve the Institute Interaction outcomes from MoU signed industries**

1. Allotted Faculty members are suggested to request more guest lectures and Industrial visits through Industrial experts from the MoU signed industries.
2. It is planned to generate more industry authored papers in future from the completed students project work carried out in the industry.
3. Planned to offer more value added credit course for the RIT students from the recent cutting edge technologies from the MoU signed industries.
4. Suggested to increase the number of outcomes from the MoU signed industries.
5. Instructed to increase research activities with MoU signed industries.

6. A.Training and Placement Progress

Mr. M. Sivagaminathan alias Balaji, AP(SG)/MECH, Placement Coordinator of Department of Mechanical Engineering

1. Presented the placement statistics for the academic year 2020 – 2021

Academic Year	Number of students opted for placement	Number of students placed	Placement Percentage
2019 - 2020	111	82	73.87%
2020 - 2021	103	99	97%

2. Listed the details of number of core, software and educational service companies visited for recruitment during 2020 – 2021 in comparison with 2019 – 2020 .

Companies	Academic Year 2019 - 2020	Academic Year 2020 - 2021
Number of core companies visited	7	19
Number of software companies visited	20	18

Companies	Academic Year 2019 - 2020	Academic Year 2020 - 2021
Number of Training & Placement companies visited	4	03
Total	31	40

3. Listed of new companies visited the campus in the academic year 2020 – 2021
    1. BYJU'S
    2. Steel Strips Wheels Limited
    3. Myunghwa Automotive Private Limited
    4. Bhavani industries
    5. East India Company
    6. SV Logistics
    7. Igarashi Motor's India Limited
    8. Axles India Limited
    9. Makolet Private Limited
    10. Gestamp Automotive Chennai Private Limited
    11. Unitech Plasto Components Private Limited
4. Presented the companies visited, position offered to students with their operating domain

S.NO	Name of the Company	Operating Domain	Designation offered to Students
1.	Rane TRW Occupant Safety Division	Production	Graduate Engineer Trainee
2.	TATA Consultancy Services Limited	Application Development and Maintenance projects	Assistant System Engineer-Trainee
3.	BYJU's	Business Development	Business Development Trainee
4.	Igarashi Motors India Limited	R&D	ACT Apperentice
5.	Steels Strips Wheels Limited	Operation	Graduate Engineer Trainee
6.	East India Private Limited	Production	Trainee Engineer
7.	Unitech Plasto Components Private Limited	Production	Graduate Engineer Trainee
8.	Bhavani Industries	Production	Trainee Engineer
9.	Myunghwa Automotive India Private Limited	Production and Quality	Graduate Engineer Trainee
10.	Sakthi Auto Components Limited	Production and Quality	Graduate Engineer Trainee
11.	Windcare India Private Limited	Design and Site Engineer	Trainee Engineer
12.	Makolet Private Limited	Digital Marketing	Digi MarketingTrainee
13.	SV Logistics	Logistics	Junior Executive

5. Presented the details of training provided to the 2017 – 2021 Batch and 2018 - 2022 students as follows

S.No.	Trainer Details	Training Details	Topics taught is training	Number of Students Attended
Training for 2017 – 2021 Batch Students				
1.	Employability Training for Service-based Companies (both Basic and Advanced batch) "Programming Skills in C" by Department of CSE	13.07.2020 to 15.08.2020	Programming in 'C' language, 1. Basic level and 2. Advanced level.	Basics – 8 Advanced – 4
2.	Mock Interview by Department of Mechanical & English, Training and placement Centre - RIT	From 14.07.2020, Weekly Twice	Both General and Technical Mock interview	53
3.	TCS Ninja Recruitment Drive - Company Specific Training program by Top freshers	01.10.2020 to 22.10.2020	All the topics related to TCS Ninja Recruitment drive.	41
4.	TCS Mock Interview for students qualified for Final round of TCS Ninja	25.11.2020	General Mock interview	1
5.	Innate Talent Limited	08.03.2021 - 13.03.2021	1. Attitude building 2. Positive thinking 3. Core industry oriented attitude building motivation, 4. Pride of engineering (in core industry), insight about core companies & work nature 5. Communication, its importance, types & its barriers & how to overcome barrier 6. Art of presentation, public speaking 7. Resume building, interview skills – basics interview skills & tips - complete process, grooming, body language, do's & don't's 8. Interpersonal skills - core industry oriented, leadership 9. Goal setting & its importance in career	43

			10. Quants / Reasoning (students preferred topic) 3. Group discussion 4. Time management & creativity	
Training for 2018 – 2022 Batch Students				
1.	Bootcamp by Department of Mathematics	25.02.2021 &26.02.2021	General Aptitude Training	98
2.	Innate Talent Limited	15.03.2021 to 20.03.2021	General Aptitude Training	98
3.	Kaar Technologies – Company Specific Training by Topfreshers	30.04.2021 to 07.05.2021	1. Number system 2. Ratio Proportion 3. Ages 4. Partnership 5. Coding decoding 6. Average 7. Alligation and Mixture 8. Direction sense 9. Syllogism 10. Percentage 11. Profit and Loss 12. Number series 13. Clocks 14. Calendar 15. Logical analogy 16. Time, Speed & Distance 17. Trains, Boat & Streams 18. Blood Relations 19. Permutations & Combinations 20. Probability 21. Time and work, pipes & Cistern, 22. Analytics reasoning 23. Seating Arrangements 24. SI&CI 25. Fill in the blanks (Grammar), prepositions, conjunction, articles, tenses, cloze passage, sentence correction, Error spotting, sentence completion, word selection 26. Reading comprehension, Parajumbles / Sentence reordering, Voices and speeches, spelling,	6

			Vocabulary (Synonyms & Antonyms)	
4.	Programming & Quantitative Aptitude Training	24.05.2021 to 03.06.2021	1. Number system 2. MNC interview coding questions 3. Percentages 4. Averages 5. Java 6. Ratio & Proportion 7. Partnership 8. SI &CI 9. Profit & Loss 10. Time & work 11. Time, Speed & Distance 12. Trains, Boats & Streams	98
5.	Python Programming Classes by CSE Department for Core branches	25.05.2021 to 02.06.2021	Python Programming	98

6. **B. Participation of Alumni in Training**

Apart from external training organisations, Alumni are also involved in providing company specific training to students. With regard to it, before each recruitment drive Alumni working in that particular company is invited for a discussion with the students appearing for the interview. The details are given below.

S.No.	Name of the Company	Name of the Alumni	Topics discussed
Training for 2017 – 2021 Batch Students			
1.	Rane TRW Occupant Safety Division	1. Mr. P. Ariramalingam 2. Ms. P. Keerthana 3. Mr. S. Shanmugham 4. Mr. M. Vignesh	The Alumni shared their interview experience and they explained the expectations of the company from the students. Further they also discussed about important technical concepts that can be asked by the HR.
2.	Infosys TNSLPP	Mr. Vikram V K	Shared his experience during the interview and listed do's and don'ts during the interview.
Training for 2018 – 2022 Batch Students			
1.	Kaar Technologies	1. Ms. Jegatha C	Shared their experience during the interview and explained the topics from which questions can be asked from.
2.	Vuram Yechnology	1. Ms. Vaishnavi Devi	Shared their experience during the interview and explained the topics from which questions can be asked from.

7. Presented the internship offers received by students through Training and Placement cell

S.No.	Name of the Company offering internship	Duration	Name of the Students
1.	Vaayusastra Aerospace Private Limited	6.5.2021 to 6.6.2021	5. Madhavakannan R 6. Raghuram K

S.No.	Name of the Company offering internship	Duration	Name of the Students
2.	Arvi Hitech Private Limited	2 months	1. Chandru K 2. Ramkumar T

8. Explained the details about NI CLAD Certificate course training program which was organized by Training and placement cell in association with Department of ECE. Five Mechanical Engineering students attended the course and two mechanical engineering students have cleared the course, the details about it is given below

S.No	Name of the Student	Marks Scored in Percentage
1.	M Vinoth	77%
2.	N Gokul Prasath	88%

9. Listed the following action plan for the ensuing academic year for the 2018 – 2022 batch Mechanical Engineering students
   - Placement training for 60 hours is planned for IV year students from July 28 for final year students.
   - **Mock Interview** will be conducted from July 28 for all the placement eligible students by Department of English and Department of Mechanical Engineering.
   - Company Specific Training will be conducted before every recruitment drive external training organisations for software companies and by alumni for core companies.
10. Enumerated the steps taken by Training and placement cell and Department of Mechanical engineering to implement suggestions given by Recruiters of the academic year 2019 – 2020
   - Students are encouraged to attend internships from second year onwards, The internship coordinator of department Mr. S. Kathiravan informs II and III year students about internship opportunities available.
   - The prime focus of Training and placement cell and Department of Mechanical Engineering is to empower students with employability skills. Different kinds and number of training programs are planned from second year till final year.
   - Skillrack platform has been used by students for practicing programming and TERV is introduced for practicing quantitative aptitude topics.
   - Company specific training is implemented for training students before each recruitment drive.
11. Highlighted the Average and Median salary of the students placed in the academic year 2020 – 2021, The details of which are given in the below mentioned table:

Academic Year 2020 - 2021	Number of Students Opted for placement	Number of students placed	Average Salary (In Rupees)	Median Salary (In Rupees)
	103	99	1,83,737	1,68,000

12. Suggestions given by PAQIC Members:

- PAQIC members appreciated the student's performance in the academic year 2020 – 2021.
- PAQIC members advised to include average and median salary in the presentation next time.
- PAQIC members also advised to concentrate on increasing the quality of placement in the academic year 2021 – 2022

6. Career Guidance cell

Presenter: Mr.P.Pavithran, AP/Mech

Mr.P.Pavithran, AP/Mech. briefed the Career guidance cell activities as follows:

1. Revealed number of events conducted during the academic year 2021 – 2022 (4 programs)

Name of the Program	Date & Time	Resource person	No of students Participated
Design Thinking-An disciplinary Domain	01.05.2021 & 10 to 11 AM	Mr.R.Pritivi Raj, Lead Ux Designer, Trimble, Chennai	103
Career Opportunities in EEE	23.05.2021 & 10 AM to 12 PM	Mr.R.Vigneswar, AP(SG)/EEE	50
Career Awareness program	14.06.2021 & 6 PM to 7 PM	Mr.P.Pavithran, AP/Mech	21
Newer and greater opportunities for students of core engineering branches	29.06.2021 & 10.30 AM to 12.30 AM	Mr.M.Anand, Principal councellor, Indian Green building Council, Hyderabad	88

2. Prepared a Career guidance manual for mechanical department.
3. Explained overall students enrolled in GATE Forum Institute, Madurai through online coaching, 17 students from 2021 and 12 students from 2022 batch, Duration of the program was 56 hours.
4. Circulated language training programs to students such as Hindi and Japanese

GATE Coaching:

5. Discussed previous year GATE Questions with all year students in WhatsApp in all days and clarifying student's doubt instantly.
6. Demonstrated that Mechanical faculties taken GATE coaching class, created a classes schedule to delivered topics such as Thermal, manufacturing, Fluid Mechanics and Production engineering.

Suggestions given by the members

6. Arrange interactive sessions between distinguished alumni who are all selected in government job sectors and pursuing students.
7. In Career guidance manual add the job specific training programs details.

7. PROFESSIONAL SOCIETIES AND TECHNICAL ASSOCIATION: PARTICIPANTS AND ITS INFERENCE

Presenter: Mr. S. Valai Ganesh, AP/Mechanical

Presented about various professional societies and club events for the AY 2020-2021

- RIT IEI Mech conducted more events than other societies/club. Planned 18 and conducted 46.
- RIT ISTE applied for Best Professional Chapter award for the AY 2019-2020
- SAEINDIA chapter participated in the following events
    - RIT CRITON has participated in SAEISS Tractor Design Competition- 2020
    - RIT SPARTANS has participated in Bicycle Design Competition- 2020.
- RIT IWS chapter made an MoU with WRI for conducting Value Added Program to the RIT students
- Totally there are 78 events planned out of which 119 events are conducted and plan versus conversion ratio is 152.56%. Conducted event details are already uploaded in college website and department newsletter

Suggestion by the members

- Suggested that number of students benefited from each societies should be included in final summary table

8. INTERNSHIP AND INPLANT TRAINING

Presenter: Mr.S.Kathiravan, AP/Mech

Mr.S.Kathiravan, AP/Mech. briefed the Internship and Inplant Training as follows:

a. Improvement in the Internship / Inplant training (2019-2020) from (2020-2021)
b. Highlighted the names of Internship from Premium Institutions like IIT Delhi, IIT Bombay.
c. Presented the Program outcomes (PO) and Programme specific outcomes (PSO) mapping with area of internship
d. Revealed the number of Internship attended by the students in MoU Signed companies.
e. Newly implemented course outcome for Internship/ Inplant training are explained.
f. Presented year wise Internship/ Inplant training, conventional practices used and evaluation techniques.
g. Internship mark statement, consolidated details and certificates has been sent to Anna University CoE office.
h. Letter has been send to include Internship in listed course in Employability enhancement courses -Anna University Centre for Academics.

Suggestions given by the members

1. The committee offered suggestions to encourage students to participate in internships and in-plant training at all of the organisations with which we have signed memorandums of understanding.
2. Asked reforms/ identifications done for academic year 21-22
3. Suggested to incorporate quantitative measures in outcome attainment.

9. EDC, IIC & NISP

Presenter: Mr.M.Santhana Maruthu Pandian, AP/Mechanical

Mr.M.Santhana Maruthu Pandian, AP/Mechanical has briefed the activities of EDC, IIC & NISP as follows:

a. The National Innovation and Startup Policy for students and faculty of Higher Education Institutions (HEIs) will enable the institutes to actively engage students, faculties and staff in innovation and entrepreneurship related activities.
b. Startup policy, thus enabling creation of a robust innovation and Start up ecosystem across all HEIs.
c. Tamil Nadu Start-up and Innovation Policy is presumed to nurture innovation, investment in R&D, infrastructure, knowledge creation, technological development and skilled manpower, resulting in high growth entrepreneurial ventures across the spectrum of sectors from agriculture, manufacturing, healthcare, education, logistics, social sector and urban development.
d. Ramco Institute of Technology a part of the Ramco group which has more than 60 years of industrial exposure is committed to fulfill to objectives of NISP and TNSIP with a contribution from its side by enabling an innovation and entrepreneurial ecosystem.
e. As a road map to this missionary vision, the RIT Innovation and Entrepreneurship Policy (RIT-IEP) is formulated under the guidance of experts from various fields including all the stake holders.
f. Presented the Mission and Vision of RIT-IEP.
g. Institution Innovation Council (IIC) of RIT was established in September, 2019. Under the guidelines of MoE Innovation Cell (MIC), various activities under the themes Innovation, Entrepreneurship, Start-ups and Intellectual Property Rights are being planned and conducted.
h. These programmes are organized and executed through various cells of RIT including RIT Entrepreneurship Development Cell (RIT ED-Cell), RIT Intellectual Property Rights Cell (RIT IPR Cell).
i. RIT IIC have secured 4 star rating for the academic year 2019 – 2020 (IIC 2.0)
j. Patents filed – 8 and copyrights registered – 8.
k. The students turned entrepreneurs are 8.

Suggestions given by the members

- Improve the number of IP portfolio.
- Encourage students to participate in more number of activities in future.

10. FACULTY PARTICIPATION AND PERFORMANCE

Presenter: Mr.N.L.Sujin, AP/Mech. and Dr.V.Sivakumar, ASCP/Mechanical

Mr.N.L.Sujin, AP/Mech. briefed the faculty participation as follows:

a. Three faculty members Mr.M.Ramar, Mr.S.Kathiravan and Mr.P.Pavithran were appointed as Assistant Professors cadre and the student faculty ratio has improved from 21.68 (2019-20) to 18.22(2020-21).

b. The faculty qualification improved from 12.75 (2019-20) to 15.5 (2020-21) by the sanction of promotion to three Assistant Professors (SG) (Dr.P.Sureshkumar, Dr.V.Sivakumar and Dr.S.Godwin Barnabas) to Associate Professors with effect from 01.01.2020.

c. There is an increase in faculty participation in ATAL FDP (34 courses) and participation from reputed institutions (IIT - 5 Programmes & NIT - 4 Programmes) by the faculty members over the period. Emphasized the effectiveness of the program participation and implementation by the faculty members.

d. The number of programs attended as a resource person by the 20 faculty members within RIT (28 programs) and outside RIT (Within the State: 20 and Other State:5 totally 25 Programs)

e. The faculty members attended 53 - FDPs, 2 - Short Term Courses, 10- Training programs, 114 - Webinars, 13-Workshops and online courses during this period.

f. The performance appraisal done with the revised pattern was discussed elaborately; identification of training needs, suggestions given by the committee members, implementation by the faculty members and its effectiveness was also discussed with two sample cases.

Dr.V.Sivakumar, ASCP/Mech. listed out the progress of the Research and Development activities as follows:

1. 31 numbers of proposals (Design & Materials – 9, Manufacturing, Thermal and Mechactronics & AI – each 5, Waste Management – 4 and Interdisciplinary – 3 (Mech.& CSE – 2 & Mech., Phy. & Che. - 1) have been submitted during 2020-2021 to various funding agencies like DST / SERB / IEI(I) / ISHRAE / TNSCST / FAER / NIAM Agri-Business Incubator

Suggestion by the members

- Enhance the number of interdisciplinary project proposal submission
- Collaborate to work with mathematical faculty and introduce mathematical modeling in the project proposal.

2. Reviewed the progress of the project "Design and Development of Dehydrating Device for Continuous Processing of Herbal Leaves" sanctioned under DST SSTP

Work completed:

- 100 % fabrication work completed
- Performance analysis was done with drying of Thulasi, Neem and Morunga leaves

Work in progress:

- The process parameters like velocity of air, drying time, electrical power input and mass of the herbal leaves are varied and experimental work is going on.
- Before and after drying the nutrition level will be tested in the laboratory

Outcome generated:

- Presented the various outcomes generated from the project in four international conferences

Suggestion by the members

- Submit a review paper and communicate one Journal paper from the outcome of the project.

3. Reviewed the progress of the sanctioned project "Industrial Automation using Smart Production Station" under AICTE MODROBS during June 2021.

Work completed:

- Installation of feeder, inspection and buffer station was over and the demonstration was given by the suppliers,

Work to be completed:

- Initiate the Phase II purchase activities

Work Plan:

- To give intensive training to the faculty members and students of RIT.

4. Reviewed the funded project "Design and Separation of Groundnut Pod Separator" under RuTAG IIT Madras and its outcomes (Product was developed and subjected to test run and a paper was presented in a National Conference)

Work completed:

- Confirmed the receipt grant: 24.08.2020
- First Review Meeting: 11.09.2020 (Online)
- Second Review Meeting:08.12.2020(Online)
- Third Review Meeting:10.02.2021(Online)
- Fourth Review Meeting: 04.03.2021(Online)
- Fifth Review Meeting: 15.07.2021(Online)

Work to be completed:

- Submit the detailed work plan to be carried out

Suggestion by the members

- Commercialize the product
- Apply patent
- Journal publication

5. Reviewed the project proposal "Dual Benefit in using the Waste Tyre for Wastewater Treatment and Production of Automotive Seals and Gaskets" under DST – WMT.

Work completed:

- Presented the content in front of Principal, Vice Principal and sponsors from industries and received feedback from them.
- Visited the municipal Waste Management Plant in Rajapalayam
- Presented the content during the final evaluation of the project and the result is awaited

Suggestion by the members

Scale up the product and measure the pressure drop in each stage and also study the physical and chemical examination properties of the water samples.

6. The status of the projects under review are SERB-SUPRA scheme - 2 (Materials and Thermal), IEI R&D Cell - 1(Design) & TNSCST student project scheme - 11 projects
7. Delivered the impact analysis and action taken for the projects submitted and Not –shortlisted / Not – recommended during this period under DST – WMT & TDT. Based upon the observations
   a. After incorporation of review comment(s), it will be communicated to the next proposal call
   b. Proposed the plan of action while submitting future proposals under Technology Development Transfer.
      i. Industry and host institute contribution in budget to be included
      ii. Proof of concept must be submitted
      iii. Patent search been done to identify the gaps
      iv. Lab or pilot study to be done
      v. Specify the TRL in the project proposal
      vi. Value addition must be there in the proposed technology
      vii. Engineering flow sheet of proposed process
      viii. SWOT analysis to be made
      ix. Road Map for commercialization

8. Listed out the following

Conference publications (Total - 37) (ICICSE-2020-Virtual Conference, IWCSET -2020, NWCCIET - 2021, ICRAME 2021 and ICETET- 2021)

Journal publications (Total - 20) (Materials Letters, Part E: Journal of Process Mechanical Engineering, Part J: Journal of Engineering Tribology, Journal of Thermal Analysis and Calorimetry, Fresenius Environmental Bulletin, Journal of Green Engineering, International Journal of computer communication and control, Journal of Rubber Research, Journal of Huazhong University of Science and Technology, Mechanics of advanced composite structures and Materials Today Proceedings)

a. Article (1) - Third Annual Technical Volume of Electronics and Telecommunication Engineering Division
b. Copyright registered (4) - i) Industrial Automation Design for Beginners ii) Energy Conservation iii) Introduction to intellectual copyright and iv) Ideation of Projects
c. Book chapters (4) - Advances in Material Research, Springer Proceedings in Materials, Springer, Lecture Notes in Networks and Systems and Green Bio composites for Biomedical Engineering
d. Status of patents (Filed - 6, Received FER - 6, Response to objections: 5 responded and 5 yet to be responded) during this period.
e. Emphasized the continuous monitoring of research activities through Department Review Meeting.
f. Revealed the research promotional activities to be done by the department for the continuous improvement in the R&D section.
g. Conduct of Extramural Lecture Series – International and National
h. Offering internship to the students from the reputed institution like IIT Madras
i. Implementation of research culture by formulating various domain heads
j. Suggested to publish a paper in WoS / SCI(E) indexing

Suggestions given by the members

1. Create the pathway to appoint Honorary Professor in the field of Mechanical Engineering
2. Offer internship to the students in the R&D section and promote research activities
3. Establish global contact and publish papers with foreign universities

11. LABORATORY UTILIZATION BASED ON OBE

Presenter: Mr.J.Jerold John Britto, AP(Sr.Gr.)/Mech. and Mr.R.Prabhakaran, AP/Mech

- Manufacturing Laboratory
  - A new virtual laboratory development under IITM PALS-NITK Scheme in progress for the following experiments
    - Lathe: Drilling and Boring; Taper Turning Operation; External thread Cutting; Internal Thread Cutting; Square head shaping; Hexagonal head shaping; Contour milling using vertical milling operation; Plain surface grinding; Cylindrical grinding;
- Thermal Laboratory
  - Project Development
    - Ongoing - "Design and Development of Dehydrating device for continuous processing of herbal leaves" DST under STTP net worth of Rs.15,75,010/-.
  - Program Organized
    - Webinar on "Virtual laboratory: Heat Transfer (Conduction, Convection & Radiation)" during NOV 06,2020 (Number of student Participants: 40)
- Fluid Mechanics Laboratory
  - Program Organized
    - Webinar on "Virtual laboratory: FLUID MECHANICS (Pump, Turbine, Flow Rate) during NOV 10,2020 (Number of Participants: 32)

- STRENGTH OF MATERIALS LABORATORY
    - Students/Faculty have utilized Impact Test machine for their Project/Research purpose (Number of student batches benefited -3).
    - Faculty have utilized Muffle Furnace for their Research purpose (Number of faculty benefited -1).
    - Students/Faculty have utilized Hardness Test Machine (Brinell & Rockwell) for their Project/Research purpose (Number of students batches benefited -4).
    - Students/Faculty have utilized Universal Testing Machine for their Project/Research purpose (Number of students batches benefited -7).
- CAD LABORATORY
    - Students have utilized ANSYS® Software packages for their Project and Competition purpose (Number of students benefited -47).
    - Students have utilized NXCAD/CREO®/Fusion 360 Software packages for their Project and Competition purpose (Number of students benefited -66).
- RIT- HARITA CENTRE OF EXCELLENCE
    - Placement Training
        - Placement Orientation programme has been conducted for all the mechanical Engineering Students by Harita Experts.
        - 17 Students were undergone online virtual training on CATIA V6.
        - 3 students have been shortlisted
- MECHATRONICS LABORATORY
    - Competition
        - Students participated India Skill Competition & won first 6 places in District Level.
    - Lab Development
        - Established "Industrial Automation using Smart Production Stations" under MODROP-RURAL Scheme net worth of Rs. 17,75,667/-.
        - One Copyrights filed on Mechatronics Laboratory Manual.
- RIT-MEDSBY INNOVATION CENTRE
        - **Product Development:** Robotic Arm Component – Final year Project; Fixture components – Third year project; Robotic parts – Arm, Base, Wrist & Elbow; Effortless Screw drive - Third year project
    - **Competition**
        - IITM PALS NEXGEN3D Hackathon - Water bottle making without supporting material – won 1st and 2nd Prize worth of Rs.3000/- & Rs. 2000/-
        - IITM PALS Innowah – Mask vending machine – Spring
        - Library Assistance Bot
        - Automated Sorting Waste in dump yard in smart cities.
    - **Consultancy**
        - Completed for SACS MAVMM Engineering College, Madurai – 5 major parts (Die, Refill Container, Juice can container) net worth of Rs. 8,850/-
    - **Program Organized**
        - Webinar on "3D Printed Robotics and its Applications" during NOV 09,2020 (Number of Participants: 22)
- DYNAMICS LABORATORY
    - Program Organized
        - Webinar on "Hands on training: Virtual laboratory (Simple Pendulum, Mass Moment of Inertia & Proell Governor) during OCT 29,.2020 (Number of Participants: 20)

Suggestions by the Expert Members

- Get sponsor from Industry (Used product for demonstration purpose)
- Planning of Additional Experiments.
- Skill enhancement programme for Lab technicians.

12. ALUMNI ASSOCIATION INTERACTIONS

Presenter: Mr.M.Lakshmanan, AP(SG)/Mech.

1. Listed out the following

a. Guest lectures conducted through Alumni

S.No.	Date	Description
1.	06.09.2020	Student Alumni Enhancing Session **"Gain Self Confidence"** by **Mr.M.Padmanapan (2015-19),** Junior Design Engineer, Harita Techserv Ltd.,TVS Motors **Mr.P.Sankar Bharathi(2015-19),** Junior Design Engineer, Harita Techserv Ltd.,TVS Motors.
2.	17.09.2020	**"Scope for Higher Studies in Abroad"** by **Mr.B.R.Harsath Samuel** (2015-19), MS Mechanical Engineering, Kaunas Technical University, Lithuania.
3.	27.09.2020	Scope in **"Oil and Petroleum Industry"** by **Mr.R.Ajith Kumar** (2014-18) Quality Engineer, Welding Inspector, ASNT Level II, Severn Glocon India Pvt. Ltd., Chennai.
4.	14.05.2021	A Webinar on **"NON-DESTRUCTIVE TESTING"** by **Mr.N.Ramakrishnan** (2014-18), NDT Inspector, Lamprell Energy Ltd., UAE.
5.	19.06.2021	Online webinar titled **"Scope in Mechanical Engineering after +2 / Diploma"** by **Ms. P. Keerthana (2015-19),** Associate Consultant, Krysalis Consultancy Services Private Limited, Chennai.
6.	11.07.2021	**"Virtual Alumni Entrepreneurs meet"**, More than 20 upcoming and current entrepreneurs attended this meeting.

a. Virtual Alumni Meet

S.No.	DATE	Description
1.	Feb 21, 2021 (Sunday) at 11.00 am	2013-2017 Batch Around 25 students actively participated.
2.	April 04,2021 (Sunday) at 11.00 am	2014-2018 Batch Around 40 students actively participated.
3.	May 30,2021(Sunday) at 11.00 am	2015-2019 Batch Around 70 students actively participated.

2. Highlighted the feedback received from the distinguished Almni

1. Additional Courses for students in the field of Mechanical Engineering

The courses given by the students are as follows:

- Diploma certification course in Fire and Safety.
- Courses in Welding, Painting and also in the field of oil and gas
- Non Destructive Testing.
- Cost Management Accounting course.
- IPR.

2. Higher Studies:

- Mr. Rajah Dharmalingam (Canada), Mr. Sibish Raghul (France) of Batch 2014-18 agreed to act as a **mentor** for our college students regarding higher studies guidance.
- Japanese, French additional language is an added advantage in all the companies and also for Higher studies and requested college to give language learning opportunities for students.

3. Training for students and Webinars:

- Motivational speech to students every two month that will enhance student skills' and they will get motivated.
- Communication skills are very important for the student to get placed.
- Creating awareness among the 3rd year and 4th year students in various job positions in the mechanical field.
- Give guidance to the juniors regarding Engineering service exams and TNPSC exams.
- Public sectors job training for students and awareness for GATE exams, etc.
- Soft skills training.
- Basics Reading of Drawings.
- Bill of Material.

4. Softwares:

- Students to be strong in Modeling, Drafting and assembly.
- Solidworks, CATIA, etc and Company specific software training should be given.
- Product lifecycle management (PLM) software.
- Programming languages like JAVA, Python.
- Microsoft Excel for data management and graph.

5. Trending Technologies:

- Automation is playing a very crucial role in recent days in companies.
- Machine Learning.

6. Research:

- Development of students research association in the department and college level.

7. Entrepreneur:

- Opportunities are high for Mechanical engineer as an Entrepreneur in Madurai, Tirunelveli regions in Molding process for that student should be strong in Design and also technically.
- Financial Management Courses should be taught to students.
- Managerial skills required addition to Technical Skills.

8. Placements:

- Categorize students according to their interest like Design, Manufacturing, Thermal, IT, etc. and training should be given.
- More Core companies can be invited for placement drives

General Feedbacks:

- Create a common platform for alumni students.
- Department is doing well and good in academics.
- Discipline learnt in our college helps us in our industry.
- Collaboration with foreign universities should be done.
- Ramco college name should be Notable.
- Gen Z students are very difficult for teachers. So, teachers need to update periodically according to them as well as in technology.

13. Listed out the support received from distinguished Alumni

a) Alumni members participated in NBA peer Team Interactive Session at 06.03.2021.
b) Acted as an Industrial Expert members for National Conference
c) Guest Lectures given to our students.
d) Placement training given to students by our Alumni Students.

e) Admission Fees and scholarship details has been sent to Alumni Groups.

14. Highlighted the ongoing process with distinguished Alumni

1. RIT Alumni Association renewal process will be carried out based on audit report.
2. Alma shine alumni management software will be introduced.
3. Alumni Newsletter (Jan 2021 to Jun 2021) preparation is going on.
4. Motivational quotes will be sent on Every Monday in Alumni groups for regular touch with alumni members.

Suggestions by PAQIC members:

a. Enhance the frequency of Alumni meet
b. Share the content discussed with the Alumni to all the current students and faculty members

13. CONTINUOUS IMPROVEMENT

Presenter: Mr.L.Karthikeyan, AP/Mech.

NBA criteria-7 Coordinators: Mr.M.Ashokkumar, AP(SG)/Mechanical
Mr.L.Karthikeyan, AP/Mechanical

Mr.L.Karthikeyan, AP/Mech. briefed the continuous improvement as follows:

1. Presented the academic audit process and actions taken thereof during the period of assessment: from 26.09.2020 to 17.07.2021 and presented external audit (Annual surveillance dated: 29.01.2021) comments for scope for improvements.
2. Discussed the actions taken by the department to address the comments given by the external audit members and its implementation & effectiveness. (Table 13.1)
3. Presented ISO form refinement (TG 01 – Annual performance appraisal form) and its effectiveness towards teaching – learning process such as improvements in innovation & Teaching (Theory & Practical), improvements in teaching performance, more tangible outcomes in project work, Curricular gap/ Content beyond the syllabus plan, improvements in activities that contribute in student success, mentoring effectiveness, R&D activities improvement and more extract from Industry interaction & MoU's
4. Presented our department faculty involvement in the refinement of ISO forms (Class committee meeting, course committee meeting & feedback form parameters with proof's
5. Presented placement index improvement in 2019-20 (74.79%) & 2020-21 (84%) and compared with last three years. (Annexure 13.2)
6. Showed the quality higher studies progression in 2019-20 (IIT's/AU/VIT etc.,) (Annexure 13.2)
7. Presented improvements in entrepreneurs 2019-20. (Annexure 13.2)
8. Placement details of 2017-21 batch is presented and compared with previous years (73% core industries / 13% service industries / 14% IT sector). The students got placed in fifteen reputed companies.
9. Presented admission promotional activities and improvements in quality admission (2021-22) to the department

Suggestions by PAQIC members:

- ✓ Record the shortfalls (from various stakeholders) in previous year and analyze the reasons
- ✓ Do the process refinement (Academic audit) with the help of HOD, senior faculty & Criteria-2,3,4 coordinators
- ✓ Constitute department level academic audit consultative committee for continuous improvements
- ✓ Review the curricular gaps and its fulfillment by content beyond the syllabus with the help of domain in charges (Design/Manufacturing/Thermal) & Criteria -2 coordinators
- ✓ Report PO's/PSO's attainment of value added courses, Extracurricular and co-curricular activities
- ✓ Analyze the scope for improvements using more innovative practices with the help of domain in charges & Criteria-2, 5 coordinators

- ✓ Identify low PO's/PSO's attainment courses and plan for corrective actions
- ✓ Analyze the high attainment courses and record good practices followed by the respective faculty members for further improvements.

Table 13.1 Audit comments and its effectiveness

Audit date	Type of audit	Scope for improvement	Actions taken for Implementation	Effectiveness
29.01.21	External audit (Annual surveillance)	Effectiveness of review process in class & laboratory delivery / Mentor system ***(Ref: Ext audit minutes Sl.No 2)***	Mentoring system is reviewed during DRM ***(Ref: DRM2/2020-21 Even Sl.No4)***	Students performance improved in skill rack performance
		Procedural part – documents incorporated in the system wherever possible ***(Ref: Ext audit minutes Sl.No 8)***	Mr.R.Arunkumar, AP/Mech created Canvas LMS for Dept. ***(Ref:https://canvas.instructure.com/courses/2149971)***	All the dept. files are kept and posted in Canvas LMS and ready for quick access
		Improvement potential – To change the structure of department ***(Ref: Ext audit minutes Sl.No 1b)***	➢ Nomination of dept. academic coordinators is done (RAK/GPR) ➢ ***(Ref: DRM5/2020-21 Even Sl.No.4)*** ➢ Nomination of dept. Placement in charge is done (MSB) ➢ Nomination of domain head to strengthen dept. activities	✓ Strengthen Teaching – Learning process ✓ Improvement in placements & easy follow up ✓ Continuous dept. growth in all fields ✓ Improvements in R&D
		Lesson plan with micro level planning ***(Ref: Ext audit minutes Sl.No 5)***	➢ To plan micro level planning ➢ Usage of LMS ➢ CE's collected by each faculty ➢ Framing dept. Project QP	✓ Improved TL process ✓ 360^0 learning environment ✓ Dynamic change

Table 13.2 Placement Index

Item	(2016-17)	(2017-18)	(2018-19)	(2019-20)	(2020-21)
Total No. of Final Year Students (N)	71	141	143	123 (111 willing)	118 (103 Willing)
No. of students placed in companies (x)	45	98	91	82	99*

No. of students admitted to higher studies with (y)	00	07	06	04*	*After Graduation*
No. of students turned entrepreneur (z)	01	02	06	06	*After Graduation*
x + y + z =	46	107	103	92	*After Graduation*
Placement Index: (x + y + z)/N	64.78	75.88	72.03	74.79	*83.89 (only X is taken)*
Average placement= (P1 + P2 + P3)/3	**70.89**			**3.9% improvement (Last 3 years)**	**13% improvement (Last 3 years)**

Higher Studies **(26.09.2020 to 17.07.2021)**

Student Name	Program to be graduated	Name of Institution joined
(2019-20)		
Mr. Gowtham G	Master of Engineering	Anna University, Chennai
Mr. Mohamed Siddiq M	Master of Engineering	Government College of Technology, Coimbatore
Mr. Upenthira I	Master of Engineering	Vellore Institute of Technology, Vellore
Mr. Arun Shenbaga Raj S	Master of Business Administration	SRM University, Chennai
(2016-17)		
Mr.Harikrishna S	Master of Engineering	Indian Institute of Technology Kharagpur

Entrepreneur (26.09.2020 to 17.07.2021)

Student Name	Type of business	Designation
(2019-20)		
Arun Shenbagaraj A	Startup: TN76 food delivery service, Tenkasi. www.tn76.in	Co-Founder
Ramar S	Family Business (Plastic items)	Owner
Thirunavukkarasan N M	Smart way - Online Shopping Business	Co-Founder
Vanniaperumal S	Textile business, Sankarankovil	Owner
Senthur R	Rajapalayam Biker's Pit shop	Owner
Venkateshwaraprasad P	Construction accessories	Owner

Prepared by **NBA Coordinator/Mech** **Reviewed by** (HOD/Mech) **Approved by** Principal